Phase 1: Flow Network Formalization

Dynamic Routing System for Underground Mine Operations

Mining-DRS Project

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Contents

1 Overview

This document formalizes the mining system implemented in the `drs_mining` codebase as a *continuous, aggregate flow network*. No individual truck objects, IDs, or stochastic travel times appear; the fleet is treated as a distributed, fluid capacity. The formalization maps directly to the codebase's `ConcentratorConfig`, `MultiFaceConcentratorController`, `ContinuousFleetLogistics`, `ContinuousMineFace`, `Stockpile`, and `ConcentratorPlant` modules.

The system is an underground base-metal flotation concentrator with:

- Two stochastic mining faces (Face 1 and Face 2), each producing ore with a different mean Ore-1 fraction.
- A fleet of LHDs (loaders) and haul trucks, continuously allocated across the faces.
- Two segregated stockpiles (Ore 1 and Ore 2) feeding a processing plant.
- A multi-mode blending controller that shifts between seven distinct operating modes.

2 Mine Topology: Directed Graph $G = (V, E)$

2.1 Node Definitions

Definition 2.1 (Graph Vertices). $V = S \cup I \cup D$ where:

$$
\begin{aligned}
S &= \{s_1, s_2\} && \text{Sources (mining faces)} && (1)\\
I &= \{n_{\text{jct}}\} && \text{Intersections (underground junction)} && (2)\\
D &= \{d_{\sigma_1}, d_{\sigma_2}, d_p\} && \text{Sinks (Ore-1 stockpile, Ore-2 stockpile, plant discharge)} && (3)
\end{aligned}
$$

Node catalogue (grounded in codebase entities):

Table 1: Graph vertices and their physical correspondences.

Symbol	**Type**	**Code Entity**	**Description**
s_1	Source	`ContinuousMineFace(face_id=1)`	Face 1 ($\bar{g}_1 = 0.85$ ore-1 fraction)
s_2	Source	`ContinuousMineFace(face_id=2)`	Face 2 ($\bar{g}_2 = 0.55$ ore-1 fraction)
n_{jct}	Intersection	`ContinuousFleetLogistics`	Underground junction / fleet merge point
d_{σ_1}	Sink	`Stockpile("Ore1Stock")`	Ore-1 stockpile
d_{σ_2}	Sink	`Stockpile("Ore2Stock")`	Ore-2 stockpile
d_p	Sink	`ConcentratorPlant`	Processing plant (final discharge)

2.2 Edge Definitions

Definition 2.2 (Graph Edges). $E = E_{\text{haul}} \cup E_{\text{route}} \cup E_{\text{process}}$ where each edge $e = (i, j)$ carries physical attributes $(L_{ij}, V_{ij}^{\max})$:

Table 2: Directed edges with physical attributes (default configuration).

Edge	**From**	**To**	L_{ij} **(km)**	$V_{ij}^{\max}$ **(km/h)**	**Category**
e_1	s_1	n_{jct}	1.5	15	Haul (Face 1 → Junction)
e_2	s_2	n_{jct}	2.2	15	Haul (Face 2 → Junction)
e_3	n_{jct}	d_{σ_1}	0	∞	Routing (logical split to Ore-1 pile)
e_4	n_{jct}	d_{σ_2}	0	∞	Routing (logical split to Ore-2 pile)
e_5	d_{σ_1}	d_p	0	∞	Process (stockpile→plant feed)
e_6	d_{σ_2}	d_p	0	∞	Process (stockpile→plant feed)

Edges e_3–e_6 are logical (zero-length); physical hauling only occurs on e_1 and e_2. Lengths come from `config.face_haul_distance = (1.5, 2.2)`. Maximum speed is `config.truck_velocity = 15` km/h.

2.3 Graph Schematic

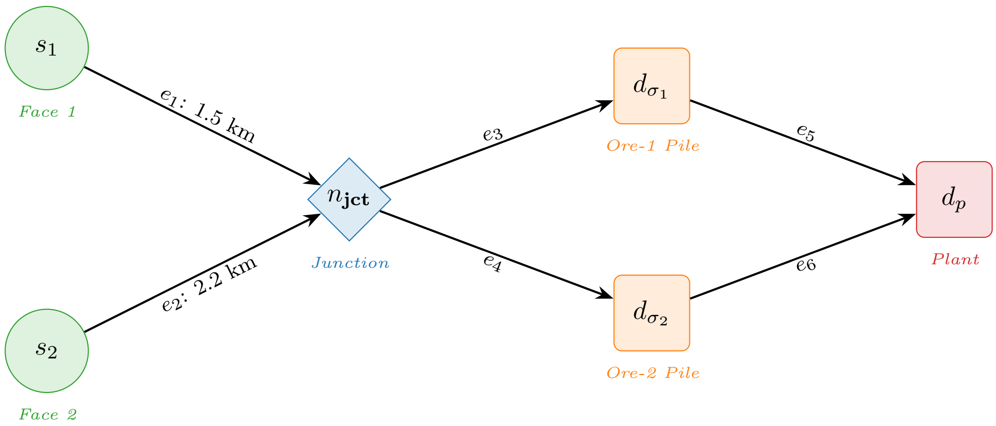

Figure 1: Directed graph $G = (V, E)$ of the baseline two-face underground mine. Sources are circles (green), the junction is a diamond (blue), stockpiles are rounded rectangles (orange), and the plant is a rounded rectangle (red).

3 Macro-State Variables

All variables are continuous functions of time t. No vehicle indices appear.

3.1 Edge Flow Variables

Definition 3.1 (Flow Rate)**.** For each edge $(i, j) \in E$, the *flow rate* $Q_{ij}(t)$ is the mass of material traversing the edge per unit time (tonnes/day).

Symbol	Edge	Description
$Q_{s_1,n}(t)$	e_1	Total extraction rate from Face 1 (maps to `face_achieved_extraction_rates[0]`)
$Q_{s_2,n}(t)$	e_2	Total extraction rate from Face 2 (maps to `face_achieved_extraction_rates[1]`)
$Q_{n,\sigma_1}(t)$	e_3	Ore-1 mass routed to Stockpile 1
$Q_{n,\sigma_2}(t)$	e_4	Ore-2 mass routed to Stockpile 2
$Q_{\sigma_1,p}(t)$	e_5	Ore-1 milling outflow from Stockpile 1 (maps to `target_stock1_outflow_rate`)
$Q_{\sigma_2,p}(t)$	e_6	Ore-2 milling outflow from Stockpile 2 (maps to `target_stock2_outflow_rate`)

3.2 Allocation Density

Definition 3.2 (Allocation Density)**.** $\rho_i(t) \in [0, 1]$ denotes the fraction of total fleet capacity allocated to face s_i. In the two-face case:

$$\rho_1(t) + \rho_2(t) = 1 \tag{4}$$

These map directly to `face_shift_allocation_fractions[i]` in the controller.

3.3 Discrete Fleet Distribution

The continuous allocation fractions ρ_i are projected onto integer equipment counts by the controller's `_distribute_discrete_fleet()`:

$$N_i^{\mathrm{LHD}}(t) = \mathrm{round}(\rho_i \cdot N_{\mathrm{total}}^{\mathrm{LHD}}) \in \{0, \ldots, N_{\max}^{\mathrm{LHD}}\} \tag{5}$$

$$N_i^{\mathrm{truck}}(t) = \mathrm{round}(\rho_i \cdot N_{\mathrm{total}}^{\mathrm{truck}}) \in \{0, \ldots, N_{\max}^{\mathrm{truck}}\} \tag{6}$$

subject to:

$$\sum_{i \in S} N_i^{\mathrm{LHD}} = N_{\mathrm{total}}^{\mathrm{LHD}}, \qquad \sum_{i \in S} N_i^{\mathrm{truck}} = N_{\mathrm{total}}^{\mathrm{truck}} \tag{7}$$

3.4 Node Process Rates

Definition 3.3 (Source Process Rate)**.** The maximum physical extraction rate at face s_i, given its current equipment assignment, is:

$$P_i(t) = R_i^{\mathrm{raw}}(t) \cdot \alpha_i \cdot \mu \tag{8}$$

where R_i^{raw} is the raw throughput from the match factor model (section 4), α_i is the face accessibility fraction, and μ is fleet mechanical availability.

Definition 3.4 (Sink Process Rates)**.** The plant acceptance rates are mode-dependent and given by:

$$C_{\sigma_1}(t) = \text{ore-1 milling rate under current mode } m(t) \tag{9}$$

$$C_{\sigma_2}(t) = \text{ore-2 milling rate under current mode } m(t) \tag{10}$$

See table 3 for all mode-specific values.

3.5 Stockpile State

Definition 3.5 (Stockpile Level)**.** Each stockpile σ_k ($k \in \{1, 2\}$) is a continuous-time integrator:

$$\dot{M}_{\sigma_k}(t) = Q_{n,\sigma_k}(t) - Q_{\sigma_k,p}(t) \tag{11}$$

with $M_{\sigma_k}(t) \geq 0$. This maps to the `Stockpile.current_mass` DRS Level.

3.6 Operating Mode State

Definition 3.6 (Operating Mode)**.** The discrete mode variable $m(t) \in \mathcal{M}$ selects from seven operating modes. Each mode fixes the target milling rates and face allocation fractions:

$$\mathcal{M} = \{\mathrm{A}, \mathrm{A}_c, \mathrm{A}_s, \mathrm{B}, \mathrm{B}_c, \mathrm{B}_s, \mathrm{SD}\} \tag{12}$$

Table 3: Operating mode parameters (from `ConcentratorConfig` defaults and `_precompute_allocations`).

Mode m	$C_{\sigma_1}^m$ (t/d)	$C_{\sigma_2}^m$ (t/d)	R_{ext}^m (t/d)	ρ_1^m	ρ_2^m
MODE_A	3600	2400	6000	0.667	0.333
MODE_A_CONTINGENCY	3900	0	3900	0.667	0.333
MODE_A_MINE_SURGING	3600	2400	$\frac{3600}{1-p}$	1.0	0.0
MODE_B	4600	800	5400	0.833	0.167
MODE_B_CONTINGENCY	0	2500	2500	0.667	0.333
MODE_B_MINE_SURGING	4600	800	$\frac{800}{p}$	0.0	1.0
SHUTDOWN	0	0	0	—	—

p = `stockpile2_routing_fraction` (dynamic Ore-2 fraction of mine output). Allocation fractions ρ_i^m are pre-computed from face generator means via $r_1 = (C_{\sigma_1} - R_{\mathrm{ext}} \cdot \bar{g}_2)/(\bar{g}_1 - \bar{g}_2)$.

4 Match Factor Model (Congestion Function)

The "congestion function" at each source is the *match factor model*, which makes effective extraction rate a non-monotonic function of fleet density. This replaces the classical BPR-style $V_{ij} = f(\rho_{ij})$ for underground mining.

4.1 Truck Cycle Time

For face s_i with haul distance L_i and N_i^{truck} trucks allocated:

$$T_i^{\text{cycle}} = \underbrace{\frac{2L_i}{V^{\max}}}_{T_{\text{travel}}} + \underbrace{t_{\text{load}} \cdot \frac{w_{\text{truck}}}{w_{\text{LHD}}}}_{T_{\text{load}}} + \underbrace{t_{\text{dump}}}_{T_{\text{dump}}} + \underbrace{\delta_{\text{traffic}} \cdot N_i^{\text{truck}}}_{T_{\text{delay}}} \tag{13}$$

where:

Symbol	Config Field	Default Value
$V^{\max}$	`truck_velocity`	15 km/h
t_{load}	`loader_cycle_time_hours`	0.0833 h (5 min)
t_{dump}	`truck_dump_time_hours`	0.033 h (2 min)
δ_{traffic}	`traffic_delay_per_truck_hours`	0.015 h/truck
w_{truck}	`truck_payload_tonnes`	30 t
w_{LHD}	`loader_payload_tonnes`	15 t

4.2 Match Factor

$$\text{MF}_i = \frac{N_i^{\text{truck}} \cdot T_{\text{load}}}{N_i^{\text{LHD}} \cdot T_i^{\text{cycle}}} \tag{14}$$

4.3 Raw Throughput

$$R_i^{\text{raw}} = \begin{cases} \dfrac{N_i^{\text{truck}}}{T_i^{\text{cycle}}} \cdot w_{\text{truck}} \cdot 24 & \text{if } \text{MF}_i < 1 \quad \text{(under-trucked: trucks dictate)} \\ \dfrac{N_i^{\text{LHD}}}{t_{\text{load}}} \cdot w_{\text{LHD}} \cdot 24 & \text{if } \text{MF}_i \geq 1 \quad \text{(over-trucked: loaders dictate)} \end{cases} \tag{15}$$

This is a *concave, piecewise-linear* function of the truck allocation N_i^{truck}, implementing a natural capacity ceiling when loaders become the bottleneck.

The final capacity at face i is $P_i = R_i^{\text{raw}} \cdot \alpha_i \cdot \mu$ as per eq. (8).

5 System Constants Dictionary

Table 4: Complete dictionary of system constants (from `ConcentratorConfig`).

Parameter	Symbol	Default	Unit
Mass Balance			
Target stockpile level	M^{target}	60,000	t
Total ore to extract	M^{total}	6,600,000	t
Warming-period extraction	M^{warm}	600,000	t

(continued)

Parameter	Symbol	Default	Unit
Critical Ore-2 level	$M_{\sigma_2}^{\text{crit}}$	20,400	t
Scheduling			
Campaign duration	T^{camp}	34	d
Shutdown duration	T^{sd}	1	d
Contingency duration	T^{cont}	1	d
Fleet shift duration	T^{shift}	0.5	d
Fleet			
Total LHDs	$N_{\text{total}}^{\text{LHD}}$	3	units
Total trucks	$N_{\text{total}}^{\text{truck}}$	10	units
Max LHDs per face	$N_{\text{max}}^{\text{LHD}}$	2	units
Max trucks per face	$N_{\text{max}}^{\text{truck}}$	6	units
Mechanical availability	μ	0.85	—
Truck payload	w_{truck}	30	t
LHD payload	w_{LHD}	15	t
Truck velocity	V^{max}	15	km/h
Loader cycle time	t_{load}	0.0833	h
Truck dump time	t_{dump}	0.033	h
Traffic delay per truck	δ_{traffic}	0.015	h/truck
Face Physical Parameters			
Face 1 haul distance	L_1	1.5	km
Face 2 haul distance	L_2	2.2	km
Face 1 accessibility	α_1	0.93	—
Face 2 accessibility	α_2	0.91	—
Face 1 mean ore-1 fraction	$\bar{g}_1$	0.85	—
Face 2 mean ore-1 fraction	$\bar{g}_2$	0.55	—
Mode-Specific Milling Rates			
Mode A: Ore-1 milling	$C_{\sigma_1}^{A}$	3,600	t/d
Mode A: Ore-2 milling	$C_{\sigma_2}^{A}$	2,400	t/d
Mode A Contingency: Ore-1	$C_{\sigma_1}^{A_c}$	3,900	t/d
Mode B: Ore-1 milling	$C_{\sigma_1}^{B}$	4,600	t/d
Mode B: Ore-2 milling	$C_{\sigma_2}^{B}$	800	t/d
Mode B Contingency: Ore-2	$C_{\sigma_2}^{B_c}$	2,500	t/d
Geological / Stochastic			
Mean ore fraction	$\bar{g}$	0.30	—
Std dev ore fraction	σ_g	0.05	—
Probability new facies	p_{new}	0.30	—
Same-facies variation	Δ_{facies}	0.01	—
Min parcel mass	$M_{\text{parcel}}^{\text{min}}$	30,000	t
Max parcel mass	$M_{\text{parcel}}^{\text{max}}$	50,000	t
Development rate per spare truck	r_{dev}	50	m/truck/d

Decision Variables Summary:

Table 5: Decision variables controlled by the DRS at each time step.

Variable	Domain	Description
$\rho_i(t)$	$[0,1]$, $\sum_i \rho_i = 1$	Face allocation fractions
$m(t)$	$\mathcal{M}$ (7 modes)	Active operating mode
$Q_{s_i,n}(t)$	$[0, P_i(t)]$	Achieved extraction rate at face i
$Q_{\sigma_k,p}(t)$	$[0, C_{\sigma_k}^{m(t)}]$	Milling outflow from stockpile k

6 Optimization Model

6.1 Objective Function: Multi-Objective

The DRS pursues a *hierarchical multi-objective*:

Primary — Maximize Total Throughput:

$$\boxed{\max_{\{\rho_i, m\}} \int_0^T \sum_{k\in\{1,2\}} Q_{\sigma_k,p}(t)\, dt} \tag{16}$$

This is equivalent to maximizing $\sum_{j\in D} Q_{v,j}$, the total flow arriving at the plant sink. In the codebase, throughput is computed as total milled mass divided by active time (excluding shutdowns).

Secondary — Minimize Blend Deviation:

$$\min_{\{\rho_i\}} \int_0^T \left(\frac{Q_{n,\sigma_1}(t)}{Q_{n,\sigma_1}(t) + Q_{n,\sigma_2}(t)} - g^{\text{target}}(m(t)) \right)^2 dt \tag{17}$$

where $g^{\text{target}}(m)$ is the mode-specific target Ore-1 fraction implied by $C_{\sigma_1}^m / (C_{\sigma_1}^m + C_{\sigma_2}^m)$.

This secondary objective is what the codebase's `_precompute_allocations()` solves analytically: given face means $\bar{g}_1, \bar{g}_2$ and target milling split $(C_{\sigma_1}, C_{\sigma_2})$, compute allocation fractions that produce the right blend:

$$\rho_1^* = \frac{C_{\sigma_1} - (C_{\sigma_1} + C_{\sigma_2}) \cdot \bar{g}_2}{\bar{g}_1 - \bar{g}_2} \cdot \frac{1}{C_{\sigma_1} + C_{\sigma_2}}, \qquad \rho_2^* = 1 - \rho_1^* \tag{18}$$

Tertiary — Maximize Mine Development:

$$\max \int_0^T r_{\text{dev}} \cdot \sum_{i\in S} \max\!\left(0,\ N_i^{\text{truck}} \cdot \left(1 - \frac{Q_{s_i,n}(t)}{P_i(t)}\right)\right) dt \tag{19}$$

Spare trucks (when $P_i > Q_{s_i,n}$) are diverted to mine development, tracked by `cumulative_mine_development`

6.2 Constraint Set

Constraint 6.1 (Conservation of Mass at Junction n_{jct})**.** Total incoming flow equals total outgoing flow at the junction:

$$\sum_{i\in S} Q_{s_i,n}(t) = Q_{n,\sigma_1}(t) + Q_{n,\sigma_2}(t) \tag{20}$$

The split is governed by the stochastic ore-1 fraction of the merged flow:

$$Q_{n,\sigma_1}(t) = \sum_{i\in S} Q_{s_i,n}(t) \cdot g_i(t), \qquad Q_{n,\sigma_2}(t) = \sum_{i\in S} Q_{s_i,n}(t) \cdot (1 - g_i(t)) \tag{21}$$

where $g_i(t)$ is the instantaneous ore-1 fraction at face i (from the facies generator, stored in `active_parcel_ore_fraction`).

Constraint 6.2 (Conservation of Mass at Stockpiles)**.** The stockpile ODE (eq. (11)) enforces:

$$\dot{M}_{\sigma_k}(t) = Q_{n,\sigma_k}(t) - Q_{\sigma_k,p}(t), \qquad M_{\sigma_k}(t) \geq 0 \quad \forall\, k \in \{1, 2\} \tag{22}$$

When $M_{\sigma_k} \leq \epsilon$, outflow is capped at inflow:

$$M_{\sigma_k}(t) \leq \epsilon \implies Q_{\sigma_k,p}(t) \leq Q_{n,\sigma_k}(t) \tag{23}$$

This is the stockout guard in `Stockpile.forward()`.

Constraint 6.3 (Source Capacity Ceiling)**.** Flow leaving each face cannot exceed its physical capacity:

$$Q_{s_i,n}(t) \leq P_i(t) = R_i^{\text{raw}}(t) \cdot \alpha_i \cdot \mu \qquad \forall\, i \in S \tag{24}$$

In the codebase: `face_achieved = min(target, real_extraction_rate)`.

Constraint 6.4 (Sink Capacity Ceiling)**.** Milling outflow is bounded by mode-dependent plant acceptance rates:

$$Q_{\sigma_k,p}(t) \leq C_{\sigma_k}^{m(t)} \qquad \forall\, k \in \{1, 2\} \tag{25}$$

Constraint 6.5 (Fleet Capacity Conservation)**.** The sum of all allocated equipment across all faces equals the total active fleet:

$$\sum_{i \in S} N_i^{\text{LHD}}(t) = N_{\text{total}}^{\text{LHD}}, \qquad \sum_{i \in S} N_i^{\text{truck}}(t) = N_{\text{total}}^{\text{truck}} \tag{26}$$

$$\sum_{i \in S} \rho_i(t) = 1 \tag{27}$$

Constraint 6.6 (Per-Face Equipment Bounds)**.** Physical limits on equipment at each face:

$$0 \leq N_i^{\text{LHD}}(t) \leq N_{\max}^{\text{LHD}}, \qquad 0 \leq N_i^{\text{truck}}(t) \leq N_{\max}^{\text{truck}} \qquad \forall\, i \in S \tag{28}$$

Constraint 6.7 (Congestion Function (Traffic Delay))**.** Edge travel velocity is a decreasing function of truck allocation density. The effective cycle time includes a linear traffic delay:

$$T_i^{\text{cycle}}(\rho) = T_{\text{travel},i} + T_{\text{load}} + T_{\text{dump}} + \delta_{\text{traffic}} \cdot N_i^{\text{truck}}(\rho) \tag{29}$$

This makes P_i (and hence $Q_{s_i,n}$) a *concave* function of ρ_i:

$$\frac{\partial^2 P_i}{\partial \rho_i^2} \leq 0 \tag{30}$$

The concavity arises because each additional truck adds a fixed traffic delay δ_{traffic} to *every* truck's cycle.

Constraint 6.8 (Campaign Timing)**.** Mode transitions are governed by campaign and contingency timers:

$$\text{Campaign ends at: } t_{\text{start}} + T^{\text{camp}} \quad (\text{or } T^{\text{sd}} \text{ for SHUTDOWN}) \tag{31}$$

$$\text{Contingency ends at: } t_{\text{cont,start}} + T^{\text{cont}} \tag{32}$$

$$\text{Fleet reallocation at: every } T^{\text{shift}} \text{ interval} \tag{33}$$

Constraint 6.9 (Extraction Target)**.** The total extraction demand is mode-dependent:

$$Q_{s_1,n}(t) + Q_{s_2,n}(t) \leq R_{\text{ext}}^{m(t)} \tag{34}$$

where $R_{\text{ext}}^m = C_{\sigma_1}^m + C_{\sigma_2}^m$ for non-surging modes. Each face's share is:

$$Q_{s_i,n}^{\text{target}}(t) = R_{\text{ext}}^{m(t)} \cdot \rho_i^{m(t)} \tag{35}$$

Constraint 6.10 (Stockout-Triggered Mode Transitions)**.** The system enforces automatic mode transitions based on stockpile depletion:

$$m = \text{MODE\_A} \;\wedge\; M_{\sigma_1} \le \epsilon \implies m \leftarrow \text{MODE\_A\_MINE\_SURGING} \tag{36}$$
$$m = \text{MODE\_A} \;\wedge\; M_{\sigma_2} \le \epsilon \implies m \leftarrow \text{MODE\_A\_CONTINGENCY} \tag{37}$$
$$m = \text{MODE\_B} \;\wedge\; M_{\sigma_2} \le \epsilon \implies m \leftarrow \text{MODE\_B\_MINE\_SURGING} \tag{38}$$
$$m = \text{MODE\_B} \;\wedge\; M_{\sigma_1} \le \epsilon \implies m \leftarrow \text{MODE\_B\_CONTINGENCY} \tag{39}$$

Surging terminates when $M_{\sigma_1} + M_{\sigma_2} \le M^{\text{target}}$.

7 Compact Model Summary

Collecting all elements, the Phase 1 optimization model is:

$$\boxed{\begin{array}{ll}
\displaystyle\max_{\{\rho_i(t),\, m(t)\}} \quad \int_0^T \big[Q_{\sigma_1,p}(t) + Q_{\sigma_2,p}(t)\big]\, dt & \\
\text{subject to:} & \\
\quad Q_{s_i,n}(t) = \min\big(R_{\text{ext}}^m \cdot \rho_i^m,\; P_i(t)\big) & \forall\, i \in S \\
\quad P_i(t) = f\big(N_i^{\text{truck}}(\rho_i), N_i^{\text{LHD}}(\rho_i), L_i, \alpha_i, \mu\big) & \text{(match factor model)} \\
\quad Q_{n,\sigma_k}(t) = \sum_i Q_{s_i,n}(t) \cdot \phi_{ik}(t) & \text{(ore routing)} \\
\quad \dot{M}_{\sigma_k}(t) = Q_{n,\sigma_k}(t) - Q_{\sigma_k,p}(t),\; M_{\sigma_k} \ge 0 & \text{(stockpile ODE)} \\
\quad Q_{\sigma_k,p}(t) \le C_{\sigma_k}^{m(t)} & \text{(plant ceiling)} \\
\quad \sum_i \rho_i(t) = 1, \quad \rho_i \in [0,1] & \text{(fleet conservation)} \\
\quad N_i^{\text{LHD}} \le N_{\max}^{\text{LHD}},\;\; N_i^{\text{truck}} \le N_{\max}^{\text{truck}} & \text{(equipment bounds)} \\
\quad m(t) \in \mathcal{M},\;\; \text{transitions per } \textbf{??}\ 6.8\textbf{??}\ 6.10 & \text{(mode FSM)}
\end{array}}$$

where $\phi_{i1}(t) = g_i(t)$ (ore-1 fraction) and $\phi_{i2}(t) = 1 - g_i(t)$.

8 Phase 1 Deliverable Checklist

✓ **Visual and mathematical graph schematic** — fig. 1 and tables 1 and 2 provide the directed graph $G = (V, E)$ with typed nodes and attributed edges.

✓ **Clean dictionary of all system constants and decision variables** — table 4 lists every constant with code mapping; table 5 lists all decision variables.

✓ **Confirmed mathematical optimization model layout** — section 6 provides the full {Objective + Constraints} formulation, summarized compactly in section 7. The model is ready for computational testing (LP/NLP solver or simulation-based evaluation).